

AWS MLA-C01 Exam Cheatsheet

Machine Learning Associate · Quick Reference

1 · METRICS & ALGORITHMS

House Price Prediction	Regression metrics → MSE / RMSE / MAE
Data Imbalance (tabular)	Optimize F1-score + apply SMOTE
Data Imbalance (images)	Optimize F1-score + augment new images
Tabular Data (default)	LightGBM (~90% of cases)
Highly interdependent + non-linear relationships	XGBoost
High-dimensional sparse data	Factorization Machines
Document topic modelling	LDA (Latent Dirichlet Allocation)
RCF (Random Cut Forest)	Unsupervised anomaly detection — NOT for binary classification

■ L1 regularization reduces unneeded features (sparsity). L2 regularization penalizes large weights — helps with generalization, not early stopping per se. Early stopping is a separate training technique.

2 · SAGEMAKER TRAINING

Custom script + SageMaker	SageMaker Script Mode (~90% of cases)
Large model > 1 GPU	FSDP (Fully Sharded Data Parallel) — splits model across GPUs
Increase no. of instances	Use SageMaker Distributed Data Parallel (SMDDP) library
Compute bottleneck	Use SMDDP library
Memory bottleneck	Spark on EMR
DDP + Python script	Check GPU utilisation with SageMaker Profiler annotations & reports
Hyperparameter tuning	Bayesian (uses past trials) + Hyperband (early stopping of bad jobs)
Will more data help? (SGD)	Use Learning Curve to decide
Opt-out training metadata	Set env var opt_out_tracking=1 in the training job

3 · SAGEMAKER INFERENCE

Same framework, multiple models	Multi-Model Endpoint (MME)
Different frameworks	Multi-Container Endpoint
Pre/infer/post in one call	Multi-Container Endpoint (serial inference pipeline)
Runs once daily, small payload	Serverless Inference — MaxConcurrency=1
Serverless, no cold start	Provisioned Concurrency → runs like real-time from first request
Cold start monitoring	CloudWatch metric: ModelSetupTime
Default auto-scaling metric	InvocationsPerInstance (use unless custom metric specified)
Variant traffic split	Weight is relative — e.g. A=3, B=1 → 75%/25% (not always 0-1)
Cost reduction, multiple models	Multi-Model Endpoint on same endpoint
Model compiler (normal models)	SageMaker Neo Compiler
Bigger / deep learning models	Deep Learning Containers (DLC)

4 · SAGEMAKER STUDIO & NOTEBOOKS

Lifecycle Configuration	Auto-install packages, clone repos, set env vars, mount EFS, auto-stop idle notebooks
Auto-stop idle notebooks	Use auto-stop-idle script from GitHub in lifecycle config
Debugging GPU/CPU/gradients	SageMaker Debugger → TensorBoard for visualisation
Pause pipeline for approval	SageMaker Callback Step
Versioning models	SageMaker Model Registry (auto-increments versions)
Organise models into groups	SageMaker Model Registry Collections
Explainability	Use simpler model (e.g. Linear Learner) for explainability; complex model for accuracy
Feature Store sync	Spark Connector (offline + online sync)
Add feature to feature group	UpdateFeatureGroup + PutRecord commands
Feature Store order	1) Create Feature Group → 2) Ingest records → 3) Access store for training
Performance steps post-dataset	1) Tune hyperparams/fine-tune → 2) Evaluate metrics → 3) Purchase Provisioned Throughput

5 · DATA & STORAGE

Preparing data for ML	AWS Data Wrangler (99%)
Non-ML ETL / data pipelines	AWS Glue
Streaming data ingestion	Kinesis Firehose (only when explicitly stated)
Raw / ingestion pipeline (default)	AWS Glue — even for raw data
ETL slow / throttled	Glue quota reached → request quota increase
Glue job on Aurora + S3 merge	Glue job on Apache Spark at set interval
Import data into Data Wrangler	Cataloged Connection (from different sources)
Semantic search on S3 docs	Kendra S3 Connector (ingest) + Kendra (query)
Low-effort RAG	Amazon Kendra Experience Builder
Detect PII (S3)	Amazon Macie
Remove PII	AWS Glue DataBrew
Detect PHI	Amazon Comprehend Medical
Network file storage (snapshots, cloning, tiering)	Amazon FSx for NetApp ONTAP
Share data without transforming	Dynamic Data Masking
Automated labeling (no humans)	SageMaker Ground Truth automated labeling

6 · NETWORKING & SECURITY

Bedrock access from private VPC/EC2	PrivateLink interface endpoint — do NOT modify outbound rules
Cross-account S3 transfer (no public IP)	S3 Gateway Endpoint + VPC Interface Endpoints for SageMaker & Redshift + update bucket policy
Comprehend model cross-account	ImportModel API (same-region accounts A → B)
S3 in VPC	S3 cannot be placed in VPC — use PrivateLink / Gateway Endpoint
Encrypt inter-container traffic	Enable intercontainer traffic encryption
Rotate API keys on schedule	AWS Secrets Manager + Lambda function
Cost tagging	Use resource tags (key-value pairs) — NOT Model Registry

7 · MLOPS & ORCHESTRATION

General / ML pipelines	Amazon SageMaker Pipelines
Multi-service orchestration (Lambda, Glue, etc.)	AWS Step Functions
Complex DAGs	Amazon MWAA (Managed Airflow) — rarely tested
CloudFormation: host a model	AWS::SageMaker::Model resource
Rolling deployment	Gradually replaces instances; batch size controls speed vs safety
Inefficient resource usage	AWS Compute Optimizer
Notebook auto-config on creation	SageMaker Notebook Lifecycle Configuration
Inference optimisation (LLM)	Speculative Decoding
Bedrock fine-tuning (no infra mgmt)	Amazon Bedrock fine-tuning jobs

8 · VISUALISATION & MONITORING

Outlier detection (chart)	Box plots or Scatter plots
Unusual small/large categories	Vertical bar chart
Feature engineering reference	• No. of bedrooms → Feature Splitting • Square footage → Log Transformation • Neighbourhood → One-Hot Encoding
Step Scaling	Adds/removes resources in varying amounts based on how far a metric exceeds a threshold

9 · ANOMALY DETECTION — THRESHOLD CHANGES

When upper/lower limits for anomaly detection are adjusted, the effect on **detection frequency** and **recall** follows this logic:

Change	Anomaly Detection Freq.	Recall
Lower threshold decreased	Decreased	Decreased
Lower threshold increased	Increased	Increased
Upper threshold increased	Decreased	Decreased
Upper threshold decreased	Increased	Increased

10 · MULTI-MODEL ENDPOINT — DEPLOYMENT STEPS

- 1 Upload existing models to the same Amazon S3 bucket path.
- 2 Create a SageMaker model with multi-model endpoints enabled.
- 3 Create an endpoint configuration referencing a multi-model container.
- 4 Deploy a real-time inference endpoint using the endpoint configuration.
- 5 Update models to use the new endpoint ID; pass model IDs to the endpoint.

11 · DOMAIN 4 — MODEL MONITOR & DRIFT TYPES

Model Monitor has 4 distinct monitor types — know exactly which to use:

Monitor Type	What it detects	Trigger keywords
Data Quality	Input feature drift — nulls, ranges, schema changes, category explosion	feature drift, schema change, data quality
Model Quality	Accuracy/metric drift — needs ground truth labels to compare predicted vs actual	accuracy drop, model degradation, ground truth
Bias Drift (ModelBiasMonitor)	Fairness drift across demographic/protected groups over time	bias over time, fairness, demographic groups

Feature Attribution Drift (ModelExplainabilityMonitor)	SHAP value changes — feature importance shifting in production	SHAP, feature attribution, explainability drift
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■ Clarify vs Model Monitor trap: Clarify = quantify bias with fairness metrics + SHAP explanations (pre/post training). Model Monitor = ongoing scheduled drift detection on deployed endpoints. Glue DataBrew is WRONG for bias — it cleans data but does not compute fairness metrics.

12 · DOMAIN 4 — MONITORING & OBSERVABILITY TOOLS

CloudWatch	Performance metrics, alarms, logs, dashboards — real-time operational monitoring. Use for endpoint latency, 5XX errors, CPU/GPU utilisation.
CloudTrail	Audit trail — WHO called WHICH API and WHEN. Use for compliance, security forensics, and triggering retraining (CloudTrail → EventBridge → Pipeline).
AWS Config	Tracks resource configuration changes over time. Use for compliance rules.
X-Ray	Distributed tracing for Lambda and API Gateway. Note: does NOT natively trace SageMaker InvokeEndpoint — trap answer.
QuickSight	BI dashboards for visualising ML metrics / business KPIs.
SageMaker Inference Recommender	Right-sizes instance type for a model BEFORE deployment. NOT for monitoring quality — that is Model Monitor.

■ Endpoint latency / 5XX errors → CloudWatch alarms on ModelLatency metric. CloudTrail + VPC Flow Logs = audit/network telemetry, NOT per-invocation latency diagnostics. CloudWatch Synthetics canary = reachability only, not SageMaker metrics.

13 · DOMAIN 4 — INFERENCE MODE SELECTION

Cost vs latency ladder: **Batch** (cheapest) → **Async** → **Serverless** → **Real-time** (always-on, most expensive)

Mode	Use when	Key limits / notes
Batch Transform	High-volume offline inference, no persistent endpoint, S3 in/out	Cheapest. Not real-time.
Async Inference	Large payloads, long processing, can tolerate queue delay	Up to 1 GB payload, 1 hr processing. Auto-scales to ZERO when idle (cost saving). NOT for real-time fraud detection.
Serverless	Spiky / unpredictable traffic, low sustained load	Has cold start. Provisioned Concurrency removes cold start.
Real-time	Sub-second latency required, always-on, bursts needing auto-scale	Up to 5 MB payload. Use for fraud detection, banking, IoT real-time.

14 · DOMAIN 4 — DEPLOYMENT STRATEGIES & EDGE

Blue/Green	Two identical environments. Switch ALL traffic at once after validation. Easy rollback.
Canary	Small % of traffic goes to new version first, then gradually expand. Safer than full cutover.
Linear	Traffic shifts in equal fixed increments over time.
Rolling	Gradually replaces old instances with new ones. Batch size = speed vs safety tradeoff.
Shadow Testing	New model receives same live traffic but results are NOT served to users. Safe pre-deployment validation with zero user impact.
A/B Testing	Both models serve real users. Use endpoint variant weights to split traffic. Weight is relative (not 0-1).
SageMaker Neo	Compile once, run anywhere — cloud AND edge devices.
SageMaker Edge Manager	Fleet management for edge devices — OTA model updates, inference monitoring on device fleet. Keywords: IoT, intermittent connectivity, offline inference, autonomous vehicles.

Spot Instances	Use for training and batch transform (fault-tolerant). Do NOT use for real-time inference endpoints — interruptions are unacceptable.
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15 · DOMAIN 4 — COST OPTIMIZATION

AWS Cost Explorer	Analyze historical spend and forecast future costs.
AWS Budgets	Set alerts BEFORE you overspend (proactive).
AWS Trusted Advisor	Flags underutilised / idle resources.
AWS Compute Optimizer	Right-sizes EC2 instances based on utilisation data.
SageMaker Inference Recommender	Right-sizes instance type specifically for ML endpoints.
Async Inference scales to zero	Auto-scales to 0 when idle — major cost saving for non-sustained workloads.
Serverless Inference	No charge when idle — good for low/unpredictable traffic.
Spot for training	Up to 90% cost saving — use managed spot training with checkpoints.
SageMaker Savings Plans	Commitment-based discount for sustained SageMaker usage.
Resource Tagging	Cost tracking & allocation per team/project — required for cost attribution.

16 · GAPS & ADDITIONS — DOMAINS 1, 2 & 3

Block specific IP	Use Network ACL (stateless, subnet-level, supports explicit deny). Security Groups are stateful and cannot explicitly deny by IP — wrong answer for IP blocking.
Lake Formation vs IAM (data lakes)	Keyword 'operationally efficient' + data lake → always Lake Formation. IAM on vast unstructured S3 data requires policy per resource — unmanageable at scale.
FSx for Lustre (ML training)	High-throughput ML training data — links directly to S3 bucket and caches it. Use for large-scale training / HPC. (Different from ONTAP which is for network file storage with snapshots.)
Data Wrangler imbalance	Also has built-in Random Oversampler / Undersampler — not just SMOTE.
Kinesis Streams vs Firehose	Streams = real-time processing with custom consumers (Lambda, KDA, custom apps). Firehose = delivery service to S3 / Redshift / OpenSearch. No processing logic.
AUC-ROC vs PR Curve	Use PR curve when class imbalance is severe. AUC-ROC can look good even when minority class performance is poor.
SageMaker Autopilot	Automated model selection + hyperparameter tuning (AutoML). Keywords: automatic algorithm selection, no ML expertise needed.
SageMaker JumpStart	Pre-built models hub, foundation models, one-click deploy. Keywords: prebuilt solution, foundation model without training from scratch.
SageMaker Experiments	Tracks training run metadata and configs. NOT for deployment governance — that is Model Registry.
SageMaker Lineage Tracking	Tracks data + model artifact lineage end-to-end. NOT for enforcing model promotion policies.
Debugger — specific rules	Detects vanishing/exploding gradients, overfitting, underutilised GPU via built-in rules.
Retraining trigger pattern	CloudTrail logs Model Monitor violation → EventBridge rule → triggers SageMaker Pipeline retraining.
Bedrock Guardrails	Filters harmful content, PII, off-topic responses in Bedrock-based apps. Separate from IAM/VPC security.
Network Isolation (SageMaker)	Blocks ALL internet and external network access from training container. Use for sensitive data.
Real-time + unpredictable bursts	Use real-time endpoint with Application Auto Scaling. Trap: Spot Fleet behind ALB is WRONG for real-time inference endpoints.

Shadow Testing vs A/B — key difference

Shadow = zero user impact (results discarded). A/B = both models serve real users simultaneously.

17 · SECURITY, ENCRYPTION & ADDITIONAL PATTERNS**Centralized IAM for multiple notebooks**

Attach ONE shared IAM role to all SageMaker notebook instances instead of individual roles. Reduces management overhead, ensures consistent permissions across instances.

Encryption in-transit during training

Enable 'encrypt inter-node training communication' in SageMaker training job config. Encrypts traffic between nodes in a distributed training cluster (not the same as S3 or EBS encryption).

Customer-managed KMS for S3 training data

Create customer-managed KMS key → encrypt S3 file using AWS Encryption SDK or CLI. Gives application-level key access control. SSE-S3 or SSE-KMS (AWS-managed) won't satisfy 'controlled application-level key access'.

Input feature distribution check

Use `emit_metrics` in SageMaker Data Quality Monitor to emit per-feature statistical metrics (mean, std, quantiles) to CloudWatch for real-time tracking.

CloudTrail log file integrity validation

Enable 'Log File Validation' in CloudTrail — creates signed digest files so you can detect if log files were modified, deleted, or forged after delivery to S3.

ETL job slow / timeouts

Most likely Glue service quota has been reached → request quota increase via Service Quotas console.

Notebook custom setup — minimal overhead

SageMaker Lifecycle Configuration → runs custom scripts automatically on instance creation/startup. No manual SSH or re-configuration needed on new instances.

Model Monitor — Quick Trigger Reference (Domain 4 reminder):

Monitor	Use when question mentions...
Data Quality	input drift, schema change, feature stats, null values, training-serving skew
Model Quality	accuracy drop, F1/RMSE worsening, ground truth labels available
Bias Drift	fairness, protected groups, demographic parity, disparate impact after deployment
Feature Attribution Drift	SHAP values changing, feature importance shifting, explainability drift

Good luck on the exam! All notes are based on observed exam patterns. Corrections are marked inline where applicable.